Horizon-AI 2026 Proposal: DeepGuard: Safe & Verifiable Agentic Neural Orchestration

Proposal ID: PROP-01-EXCELLENT | **PI:** Dr. Elena Vance
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Requested Grant: €2,400,000.00

Abstract

DeepGuard establishes a formal verification foundation for multi-agent deep learning systems in high-reliability operations. Our framework combines symbolic verification guarantees with state-of-the-art transformer agents.

Scientific Methodology

We formalize agent transitions via neural abstraction layers. Benchmark evaluations on the SafeAgent-Bench suite confirm consistent execution within mathematical bounds with zero safety invariant violations.

Impact And Dissemination

Open-source release of the DeepGuard engine with Apache 2.0 license. Direct integration pilot planned with 2 European aerospace industrial partners.

Work Packages

WP1: Symbolic abstractions. WP2: Multi-agent protocol. WP3: Pilot industrial deployment. WP4: Dissemination.

Budget

Total direct personnel is €1.40M covering 120 person-months across 3 institutions. Subcontracting is strictly €200k (8.3%) for specialized formal audit. Indirect costs are exactly 25% of eligible direct costs.